Sharan Teja Medikar

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer with practical experience across LLM evaluation and refinement, agentic AI, machine learning, Computer Vision and Vision Language Models, supported by measurable experimentation and real-world software delivery. Built an execution feedback code generation system that improved Qwen2.5-Coder MBPP pass@1 from 71.9% to 82.44%, and led operational software from stakeholder approval to use across approximately 30 patrol vehicles and 60+ users.

TECHNICAL SKILLS

AI / ML: Large Language Models, Generative AI, AI Agents, Agentic Workflows, Machine Learning, Deep Learning, Computer Vision, Vision Language Models, CLIP, LLM Inference, Model Evaluation, Benchmarking, Prompt Engineering, Prompt Tuning, Fine Tuning / Model Adaptation, LoRA, Transfer Learning, RAG, Failure & Error Analysis, Deterministic Workflow Logic

Frameworks / Data: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SQL

Software Engineering: Python, JavaScript, TypeScript, Full Stack Development, APIs, React, Express.js, Firebase, SQLite, Playwright, Automated Testing, Browser Automation, Git, Linux, Bash

EDUCATION

MSc Artificial Intelligence | University of Surrey

Sep 2025 - Sep 2026

BTech Computer Science and Engineering | National Institute of Technology Silchar

Jun 2021 - May 2025

PROJECTS

Self-Refining Code Generation Using Execution Feedback

MSc Artificial Intelligence Dissertation | University of Surrey

- Built an execution feedback refinement system in Python and PyTorch that generated, executed and iteratively repaired candidate programs using Qwen2.5-Coder-7B-Instruct and CodeLlama-13B-Instruct.
- Compared single pass generation, compute matched Best-of-5, adaptive refinement and fixed k refinement across HumanEval, MBPP, HumanEval Pro and MBPP Pro, using execution derived feedback and deterministic convergence/stopping logic.
- Improved Qwen2.5-Coder MBPP pass@1 from 71.9% to 82.44% with adaptive refinement; HumanEval Pro performance improved from 65.24% to 70.12%.
- Analysed failure modes, regressions and convergence to identify when iterative refinement improved or degraded code generation accuracy.

AI Job Application Agent

Personal / Applied AI Engineering Project | Python, Playwright, SQLite

- Built an AI assisted job search agent combining vacancy discovery, deterministic eligibility filtering, semantic matching/scoring, evidence grounded application package generation and local LLM processing.
- Integrated Playwright browser interaction with SQLite backed persistent workflow state, including browser based ATS interaction and duplicate prevention.
- Implemented schema/structured output validation, automated tests, bounded retry/deadline handling, human approval gates and deterministic safeguards for ambiguous or sensitive decisions.

Vision Language Model Prompt Learning & Adaptation

CLIP | Prompt / Parameter Efficient Adaptation | Model Evaluation

- Evaluated and compared prompt and parameter efficient adaptation methods including CoOp, CoCoOp, KgCoOp, CuPL, PromptSRC, ProGrad, LoRA, Tip-Adapter, CLIP-Adapter and MaPLe across six image classification datasets.
- Improved EuroSAT performance from 53.99% zero shot to 89.70% using model adaptation and achieved 91.23% Top 1 accuracy on Oxford Flowers with CoOp.

Physics-Informed Environmental Machine Learning

AI & Sustainability

- Built a physics informed LSTM over 28,632 site year observations spanning 26,446 locations and compared it with a standard LSTM and XGBoost using temporal testing.
- On the 2024 temporal test, achieved RMSE 32.053 and R^2 0.291, reducing RMSE by 34.5% versus XGBoost and 9.7% versus a standard LSTM.
- Included interpretability and environmental equity analysis and reduced model size from approximately 1.8 MB to 908 KB using float16.

AI & Health | ELSA Mental Health Prediction

- Built and evaluated supervised machine learning models on 10,838 ELSA participants using Python, scikit-learn and 5-fold cross-validation.
- Gradient Boosting was the best performing evaluated model, achieving an AUC of 0.853; the Wave 1 to Wave 3 longitudinal cohort contained 6,511 participants.

EXPERIENCE

Founder & Technical Lead | Police Night Patrolling System

Operational software project with Government of Assam stakeholders | Dec 2023 - Sep 2024

- Originated and pitched a real time police night patrolling platform, secured district government stakeholder approval and assembled a five person team.
- Worked directly with government stakeholders through development, testing, user acceptance and technical handover, contributing to authentication, backend and real time application functionality using Flutter, Firebase and Express.js.
- System reached operational use across approximately 30 patrol vehicles and 60+ users.

Software Engineering Intern | Fleckor Tech

Hyderabad, India | Jun 2024 - Jul 2024

- Contributed frontend and backend components across 3+ core product features and API integrations within an existing development environment.
- Resolved approximately 15-20 software defects within sprint timelines.

Freelance Software Developer

- Delivered approximately 25-30 websites and 10-12 applications for clients and small businesses, including complete builds and modifications to existing software.
- Worked across frontend, backend, full stack and UI/UX development.